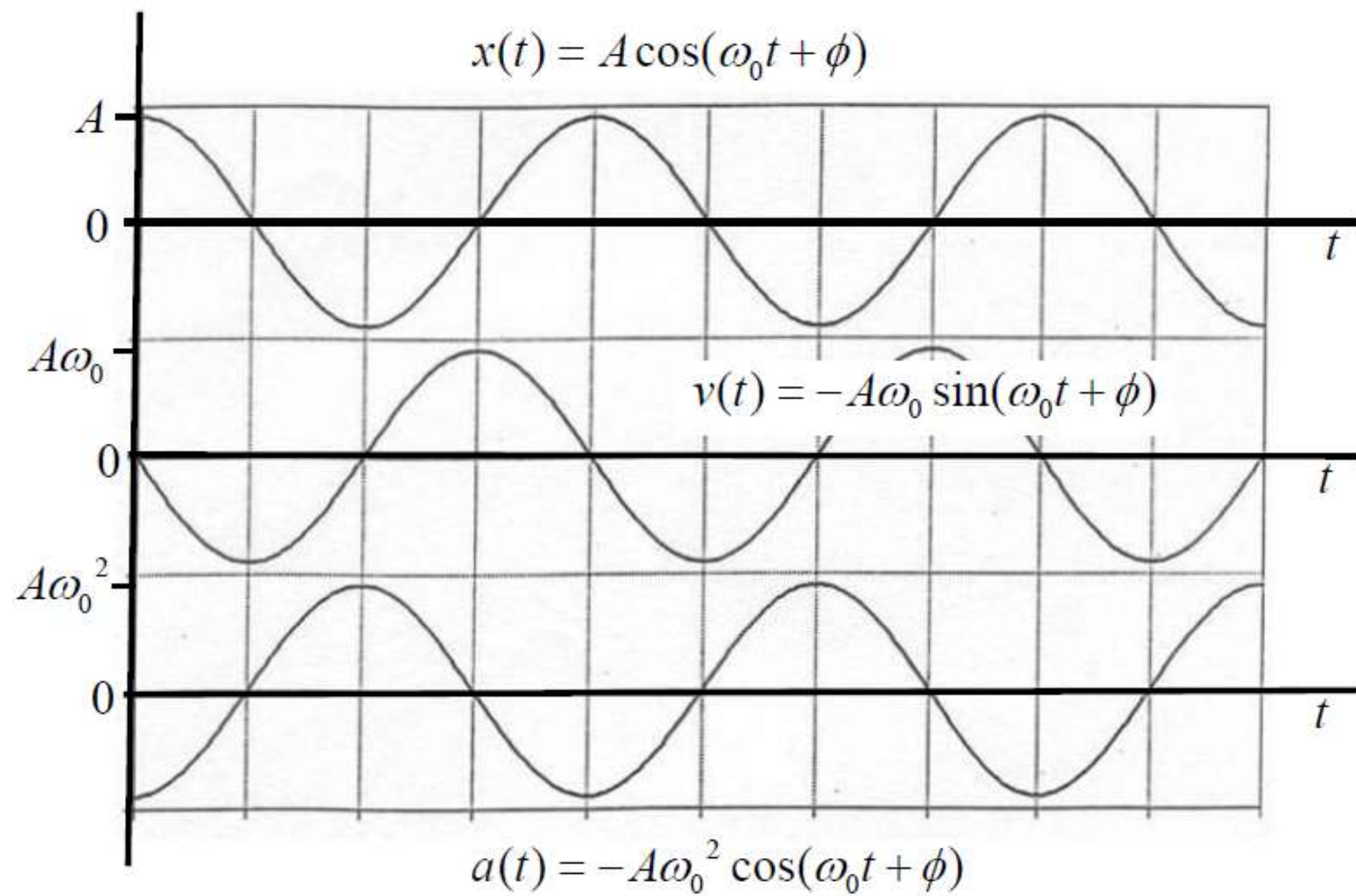
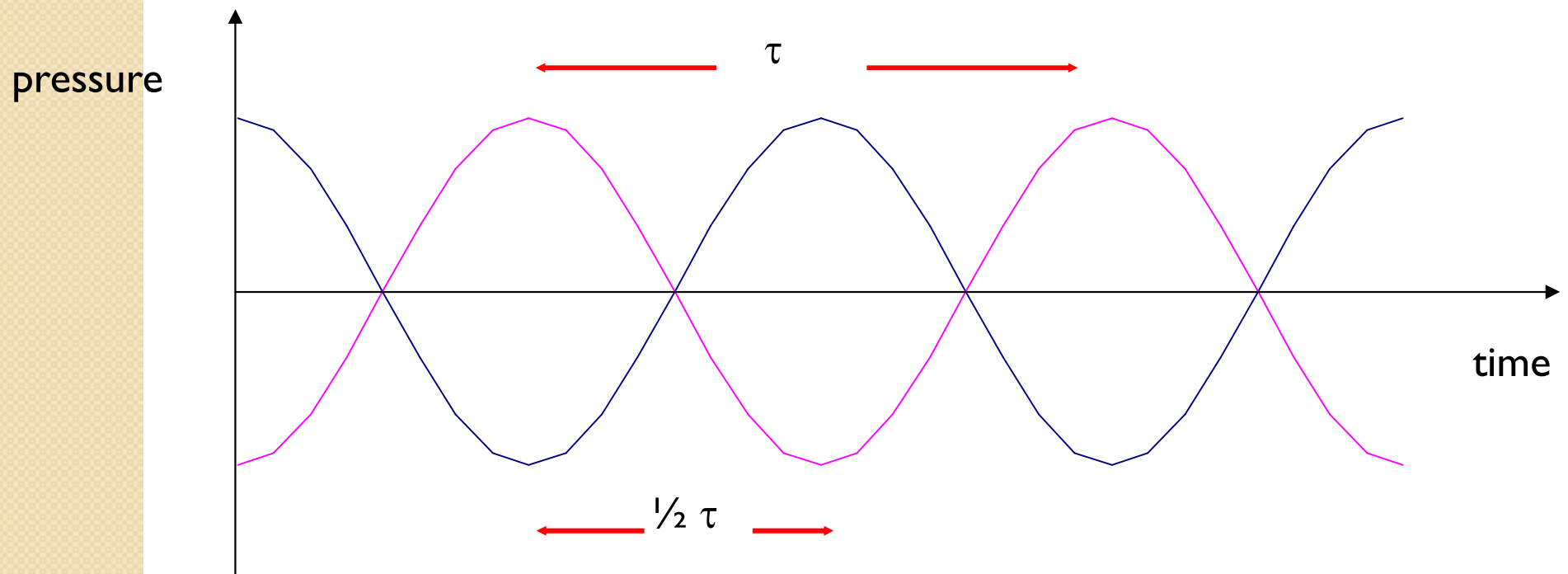




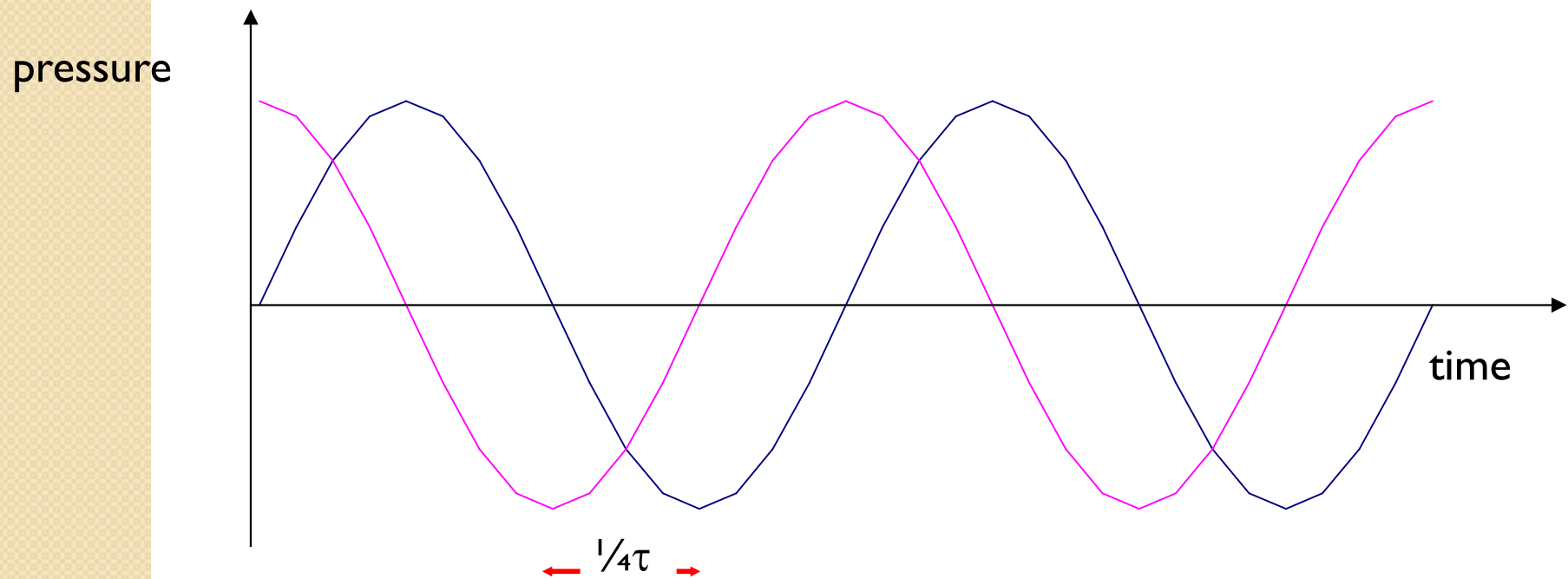
Phase Difference

- The phase of periodic wave describes where the wave is in its cycle
- Phase difference is used to describe the phase position of one wave relative to another

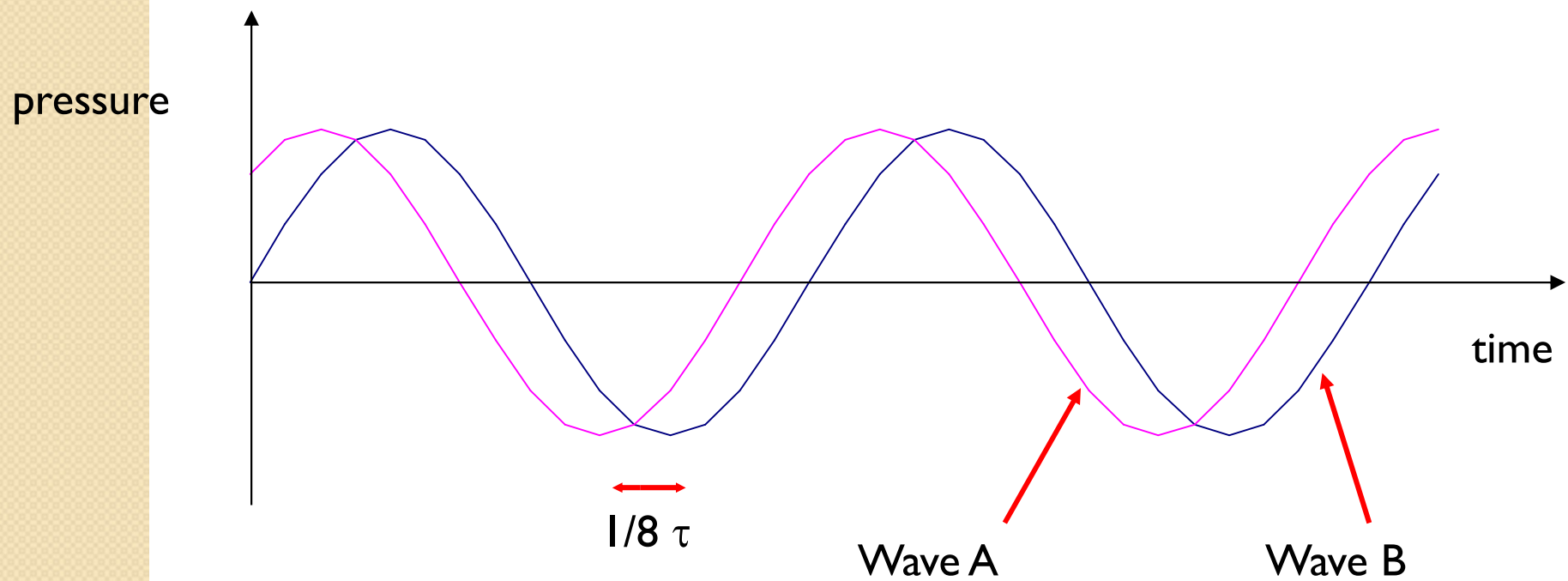
Phase Difference 180°



Phase Difference 90°



Phase Difference 45°



Simple harmonic oscillator

At $t = 0$, write $x = x_0$ and $v = v_0$.

Then at $t = 0$:

$$\left. \begin{aligned} x_0 &= A \cos(\phi) \\ v_0 &= -\omega_0 A \sin(\phi) \end{aligned} \right\} \tan \phi = -\frac{v_0}{\omega_0 x_0}$$

$$\dots \text{ and } x_0^2 + \left(\frac{v_0}{\omega_0} \right)^2 = A^2 \cos^2(\phi) + A^2 \sin^2(\phi) = A^2$$

$$\therefore A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_0} \right)^2}$$

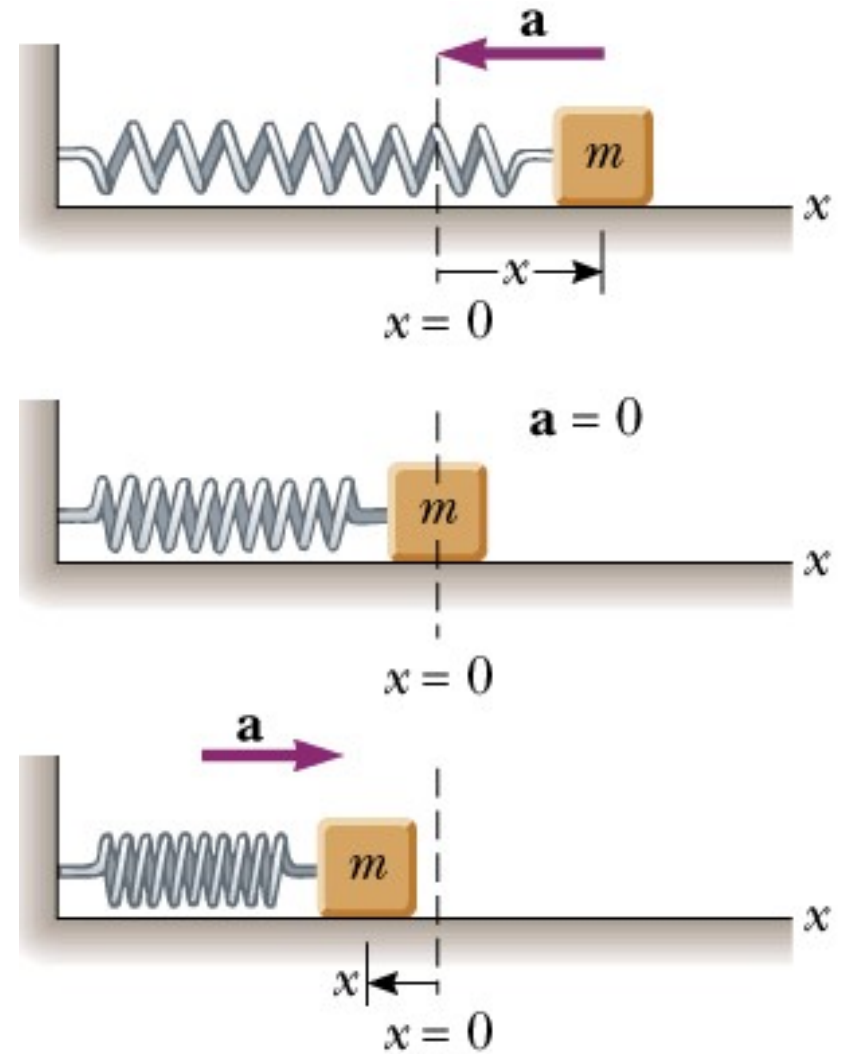
Ex. A block of mass 680g is fastened to a spring of spring constant 65N/m . The block is pulled a distance 11cm from its equilibrium on a frictionless table and released

- (a) What are the angular frequency, the frequency, and the period of the motion?
- (b) What is amplitude of the motion?
- (c) What is the maximum speed of the block?

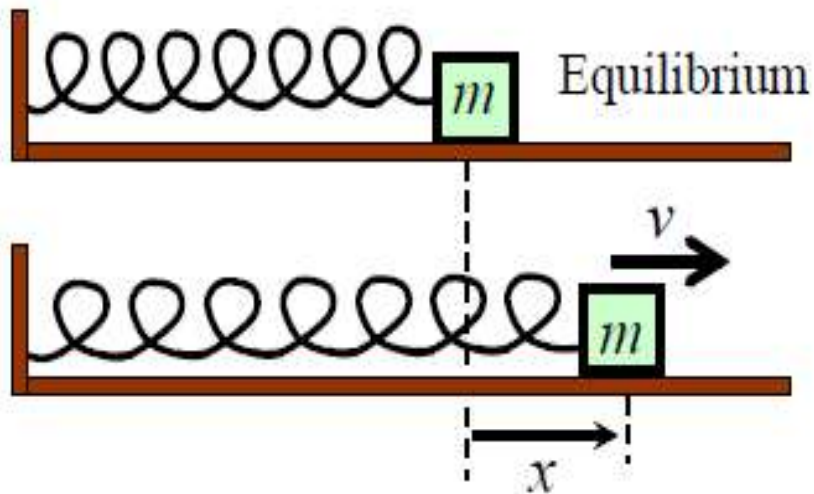
Example

A spring stretches by 3.90 cm when a 10.0 g mass is hung from it. A 25.0 g mass attached to this spring oscillates in simple harmonic motion.

- (a) Calculate the period of the motion.
- (b) Calculate frequency and the angular velocity of the motion.



Mass-spring oscillator: an energy approach



Suppose that the mass has a speed v when it has displacement x

$$\text{Kinetic energy of mass} = \frac{1}{2}mv^2$$

$$\text{Potential energy of spring} = \int_0^x F dx' = \int_0^x kx' dx' = \frac{1}{2}kx^2$$

There are no dissipative mechanisms in our model (no friction).
... the total energy of the mass-spring system is conserved.

$$\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{constant}$$

Mass-spring oscillator: an energy approach ...2

For our mass-spring system: $\frac{1}{2}mv^2 + \frac{1}{2}kx^2 = \text{constant}$

$$\therefore \frac{d}{dt} \left(\frac{1}{2}mv^2 + \frac{1}{2}kx^2 \right) = 0$$

$$\therefore mv \frac{dv}{dt} + kx \frac{dx}{dt} = 0$$

$$\therefore mv \frac{dv}{dt} + kxv = 0$$

$$\therefore m \frac{dv}{dt} + kx = 0$$

$$\therefore \frac{d^2x}{dt^2} = -\frac{k}{m}x$$

... as before

Mass-spring oscillator: an energy approach ...3

For the mass-spring system: $x = A \cos(\omega_0 t + \phi)$

$$\text{Potential energy} = \frac{1}{2} k x^2 = \frac{1}{2} k A^2 \cos^2(\omega_0 t + \phi)$$

$$\text{k.e.} = \frac{1}{2} m v^2 = \frac{1}{2} m [-A \omega_0 \sin(\omega_0 t + \phi)]^2 = \frac{1}{2} m A^2 \omega_0^2 \sin^2(\omega_0 t + \phi)$$

$$\text{Total energy} = \text{p.e.} + \text{k.e.}$$

$$= \frac{1}{2} k A^2 \cos^2(\omega_0 t + \phi) + \frac{1}{2} m A^2 \omega_0^2 \sin^2(\omega_0 t + \phi)$$

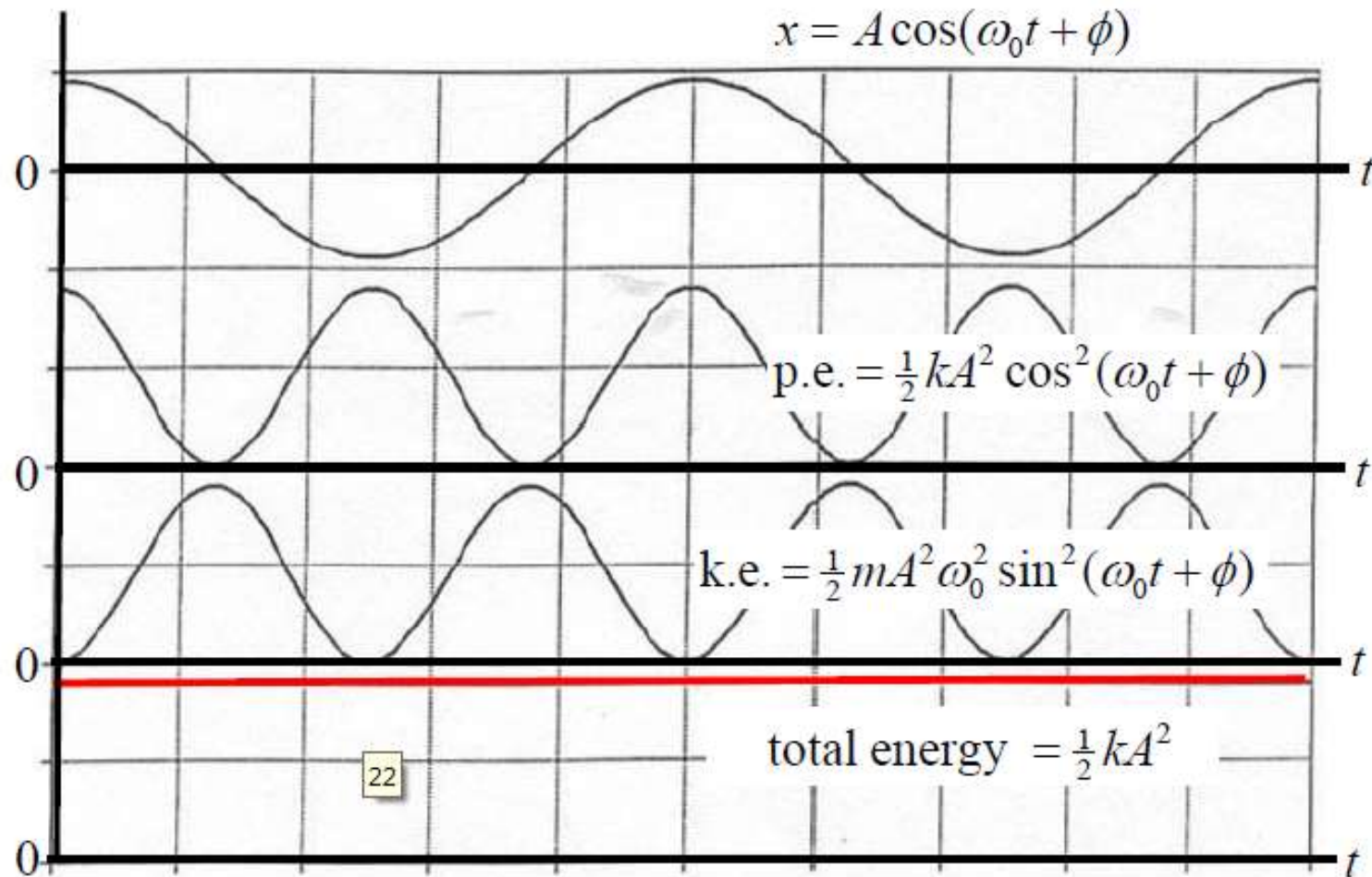
$$= \frac{1}{2} k A^2 \quad (= \frac{1}{2} m \omega_0^2 A^2) \quad (\because E \propto A^2)$$

$$\text{We can now write:} \quad \frac{1}{2} k x^2 + \frac{1}{2} m v^2 = \frac{1}{2} k A^2$$

$$\therefore v = \pm \sqrt{\frac{k}{m} (A^2 - x^2)} \quad \text{or} \quad v(x) = \pm \omega_0 \sqrt{A^2 - x^2}$$

(useful)

Energy of the mass-spring simple harmonic oscillator



K.E and P.E of SHM

